MIL-T-19229B(AS)
31 DECEMBER 1969
SUPERSEDING
MIL-T-19229A(Aer)
8 May 1958

MILITARY SPECIFICATION

TRANSMITTER, ANGLE OF ATTACK OR SIDESLIP LOCAL, MS28066

This specification has been approved by the Naval Air Systems Command, Department of the Navy

1 SCOPE

1.1 Scope - This specification covers design and all performance requirements for procurement of angle of attack or sideslip transmitters, for use with an indicator flight data computers armament computers and approach power control systems The configuration of each transmitter shall be governed by the appropriate dash number of MS28066

2 APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein

SPECIFICATIONS

Military

MIL-D-1000	Drawing Engineering and Associated Data
MIL-C-5015	Connectors Electric, AN Type
MIL-E-5400	Electronic Equipment Aircraft, General Specification for
MIL-S-8516	Sealing Compound, Synthetic Rubber, Electric Connectors and Electric Systems, Accelerator Required

FSC 6610

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SPECIFICATIONS

Military (Continued)

MIL-T-18303	Test Procedures, Preproduction, Acceptance, and Life for Aircraft Electronic Equipment, Format for
MIL-I-18856	Indicator, Angle of Attack
MIL-S-20708	Synchros, 60 and 400 Cycle, General Specification
MIL-C-23866	Control Set, Approach Power, AN/ASN-54 (V)
MIL-C-26482	Connector, Electric, Circular, Miniature, Quick Disconnect
MIL-I-81400	Instruments Aircraft, General Specification for

STANDARDS

Military

MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-470	Maintainability Program Requirements (for Systems and Equipments)
MIL-STD-471	Maintainability Demonstration
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-785	Requirements for Reliability Program
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MS25185	Connector, Plug, Electric, For Potting
MS28066	Transmitter, Angle of Attack or Sideslip, Local

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2.2 Availability of Documents - When requesting any of the applicable documents, refer to both title and number. Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 1051, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

3. REQUIREMENTS

3.1 Qualification - This specification makes provision for qualification testing. Production transmitter samples shall pass all tests specified herein.

3.2 General Design Requirements - Selection and use of parts and materials and general requirements for design and construction shall be in accordance with the requirements of MIL-I-81400.

3.3 Selection of Government Documents - Specifications and standards for necessary commodities and services not specified herein or in MIL-I-81400 shall be selected in accordance with MIL-STD-143.

3.4 Weight - The total weight of the transmitter shall not exceed that specified in MS28066 for each dash number

3.5 Performance -

3.5.1 Reliability - The supplier shall establish a reliability assurance program in accordance with MIL-STD-785.

3.5.1.1 Operational Stability - The transmitter shall operate with satisfactory performance, continuously or intermittently for a period of at least 1,500 hours without the necessity for readjustment or relubrication.

3.5.2 Longevity - The transmitter shall have a least operating life of 5000 hours with reasonable servicing and replacement of parts Parts requiring scheduled replacement shall be specified by the contractor.

3.5.3 Reliability in Mean Time Between Failure (MTBF) - The transmitter shall have 1,500 hours of mean (operating) time between failures when tested and accepted as outlined under the Reliability Assurance Tests requirements.

3.6 Electrical Requirements -

3.6.1 Cables - The transmitter shall provide for the use of cables in accordance with MIL-E-5400. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. The attached cable and connector attached to the cable shall be supplied as part of the transmitter as indicated in MS28066.

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3.6.2 Interference Control - The generation of radio interference by the transmitter and the vulnerability of the transmitter to radio interference shall be controlled within the limits of MIL-STD-461 for Class ID equipment.

3.6.3 Service Conditions - The transmitter shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-E-5400 for Class 2 equipment to the high operating temperature of 190° C, except as modified herein.

3.6.4 Primary Input Power Requirements - The transmitter shall give specified performance from any of the following power sources with characteristics as defined in MIL-STD-704 having limits as specified therein The power required shall not exceed the amount designated in MS28066

AC Power 115v, (Single Phase) Category B 60 or 400 Hz
DC Power 28v, Category B
AC Power 26±1 v, (Single Phase) 400 ± 20 Hz for synchro

3.7 Manufacturer's Part Number - Changes in manufacturer's numbers shall be governed by the drawing number requirements of MIL-D-1000/1

3.8 Maintainability - The contractor's maintainability program shall be in accordance with MIL-STD-470

3.9 Detail Requirements - The angle of attack transmitter configured to MS28066, shall provide proportional electrical outputs for use with an air data computer, an approach power control set conforming to MIL-C-23866, and an indicator conforming to MIL-I-18856 to provide indication of the local angle of attack or sideslip of the aircraft The transmitter shall be designed so that, above the minimum operative air speed of the transmitter, the mechanism of the transmitter is continuously oriented by the sensor probe by the local airflow.

The surface hardness of the probe shall exceed 30 Rockwell C scale The probe shall be mechanically coupled to drive electrically independent output components either potentiometers or synchros or any combination thereof. Various output component combinations, range of angular travel and probe to output shaft ratio shall be in accordance with MS28066. Potentiometers shall be designed to dissipate 2 watts of power continuously while the transmitter is located in an ambient temperature of approximately 25° C. Each wiper circuit shall be capable of 30 milliamperes at 55° C Potentiometer total resistance change due to temperature shall not exceed 700 parts per million per degree C Synchros shall meet the requirements of MIL-S-20708. The design of the transmitter shall be reasonably simple to permit overhaul or repair with the minimum use of special tools and fixtures. The transmitter shall be so constructed as to withstand without failure the normal strains of jars, vibrations and such other conditions as are incident to shipping, storage, installation and service.

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3.9.1 Filter - A suitable filter shall be provided to prevent foreign matter from entering the pressure balancing chamber. Drain holes shall be provided to eliminate moisture from the probe before it can enter the chamber.

3.9.2 Probe Alignment - Dowel pins shall be provided as specified on MS28066 to permit accurate alignment of the transmitter with a fixed reference line on an aircraft.

3.9.3 Case - The outline dimensions of the transmitter case shall conform to MS28066. The case shall be made of nonmagnetic low density metal with a uniform texture. The case shall be finished with a protective coating which will not crack, chip, or scale during normal service life or due to extremes of atmospheric conditions. The transmitter shall withstand a differential atmospheric pressure of 15 pounds per square inch absolute without degraded performance.

3.9.4 Wiring - Internal wiring shall be color coded, neat and accomplished in such a manner that individual wires may be easily traced. Wires shall be secured mechanically to the terminals prior to soldering.

3.9.5 Electrical Connector - The electrical connector used shall conform to MIL-C-5015 or MIL-C-26482 as designated in MS28066. An approved potting method shall be employed to provide for moisture proofing of the connector and support of the wires. The sealant material shall be in accordance with MIL-S-8516. Connectors for this purpose shall be equivalent to MS25183.

3.9.6 Heating Elements - The heating elements of the airstream direction probe and case shall operate at 28V DC or 115V AC as specified in MS28066. The probe heating element shall provide adequate de-icing and anti-icing under all service conditions (including still air operation) by maintaining the probe surface temperature between 110° C and 140° C. The case heater shall maintain the case air temperature between 30° C and 55° C to prevent condensation.

3.9.7 Deflection of Probe - The transmitter shall be capable of supporting a 150 pound load at the tip of the probe without any permanent deformation or misalignment.

3.9.8 Interchangeability of Reordered Transmitters - Interchangeability shall exist in transmitter reorders between units and all replaceable assemblies, subassemblies, and parts of the supplier's stipulated transmitter previously supplied unless otherwise contractually specified. The new supplier's transmitters shall be interchangeable with any previous stipulated transmitter in fit, form, function, and system interface compatibility without selection or alteration.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of

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all inspections and tests as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein acceptable to the Government The Government reserves the right to perform any of the tests set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements

4.2 Classification of Inspections - Items covered by this specification shall be subjected to the following inspections to determine compliance with all applicable requirements.

(1) Qualification Inspection Tests performed on sample transmitters submitted for approval as a qualified product.

(2) Quality Conformance Inspection· Tests performed on transmitters manufactured and submitted for acceptance under contract.

4.3 Qualification Inspection - The qualification inspection of the transmitter shall consist of all of the examinations and tests of this specification performed in the order specified under the paragraph headed Test Methods and shall include Reliability Qualification Phase and Maintainability Demonstration Tests.

4.3.1 Qualification Inspection Sample shall consist of a minimum of five transmitters manufactured in accordance with this specification The transmitters submitted for qualification inspection shall have been previously subjected only to the Individual and Sampling Plan A Inspections. Three samples shall be forwarded with detailed quantitative test data for all tests required by this specification at the supplier's expense to the laboratory designated in the letter of Authorization. The supplier shall conduct Reliability and Maintainability Tests on the other samples at his plant under government surveillance.

4.3.1.1 Qualification Inspection Sample Identification - The qualification inspection samples shall be plainly identified by durable tags securely attached, and marked with the following information

Sample for Qualification Inspection
Transmitter Angle of Attack or Sideslip Local MS28066-(__)
Submitted by (Manufacturer's name date and part number).
for Qualification Inspection in accordance
with Specification MIL-T-19229B(AS) under authorization
(reference letter authorizing tests)

4.4 Quality Conformance Inspection - The supplier shall furnish all samples and shall be responsible for accomplishing all the inspections and tests as designated by the procuring activity. Quality conformance inspection shall be under the supervision of the government quality control

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representative. The supplier shall furnish test reports showing detailed quantitative results for all tests required by this specification, signed by an authorized representative of the supplier or laboratory as applicable. Acceptance or approval of material during the course of manufacture shall in no case be construed as a guarantee of the acceptance of the finished product. Quality Conformance Inspection shall consist of the following tests.

(a) Individual Tests
(b) Sampling Plan A Tests
(c) Sampling Plan B Tests
(d) Reliability Assurance Tests
(e) Longevity Tests
(f) Maintainability Demonstration Test

4.4.1 Individual Tests - Each transmitter submitted for acceptance shall be subjected to the Individual Tests. These tests shall determine compliance with the requirements of material, workmanship, and operational accuracy. As a minimum, each transmitter accepted shall have passed the following test methods in the order listed.

Examination of Product
Heater Power
Mechanical Travel
Potentiometer Calibration
Potentiometer Resistance
Potentiometer Noise
Dielectric Strength
Aerodynamic Alignment
Static Friction
Synchro Accuracy

4.4.2 Sampling Plans - The Sampling Plans shall consist of Sampling Plan A and Sampling Plan B tests. The test samples selected for sampling tests shall first have passed the Individual tests. The test samples which have been subjected to Sampling Plan A tests shall not be delivered on contract until they have been refurbished and resubmitted and passed all the Individual tests. Test samples which have been subjected to the Sampling Plan B tests shall not be delivered on contract.

Quantity Offered for Acceptance	Quantity to be Selected for Sampling Plan A Tests
First 15	1 (Zero when Sampling Plan B is invoked)
Next 50	1
Next 75	1
Next 100	1
Each additional 200 or fraction thereof	1

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When a defective transmitter is detected, no items from those still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. In addition, when a failure occurs, shift to one sample out of the next fifteen and proceed as indicated (One from next 50, etc.)

4.4.2.1 Scope of Sampling Plan A Test - Each sample selected for Sampling Plan A Tests shall be subjected to the following test methods in the order listed.

Individual Tests
Magnetic Effect
Potentiometer Linearity
Heating Element Endurance
Damping
Time Constant
Low Speed Operation
Vibration Error and Failure
Pressure

4.4.2.2 Sampling Plan B Test - Two transmitters shall be selected at random from the first 15 produced on contract and submitted within 10 days after manufacture. The samples shall be forwarded at the supplier's expense to a government laboratory designated by the procuring activity. Each sample shall be plainly identified by a durable tag, securely attached and marked with the following information:

Transmitter, Angle of Attack or Sideslip, Local MS28066-(__)
Submitted by (Manufacturer's name, date) for Production Acceptance Sampling Plan B Test of MIL-T-19229B(AS) in accordance with Contract/Order No _____
Manufacturer's Part No _____

4.4.2.2.1 Sampling Plan B Approval - Approval of Sampling Plan B transmitters shall be by the procuring activity upon satisfactory completion of the designated tests. Any design, material or performance defect made evident during these tests shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the sample units to pass any of the tests shall be cause for deliveries of transmitters under the contract to cease until proper corrective action is approved and accomplished.

4.4.2.2.2 Scope of Sampling Plan B Tests - Each sample selected for Sampling Plan B Tests shall be subjected to the following test methods in the order listed.

Sampling Plan A tests
Electromagnetic Interference
Torque
Deflection of Probe

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De-Icing
Low Temperature Exposure
High Temperature Exposure
Shock
Endurance
Humidity
Fungus Resistance
Salt Fog
Dust
Rain
Internal Examination

4.5 Standard Conditions for Tests - Unless otherwise specified, all tests required by this specification shall be made under the following conditions

Temperature	Room ambient 25 ±5° C
Pressure	Normal atmospheric (approximately 29.92 inches Hg.)
Humidity	Room ambient 40 to 90% relative humidity
Input Voltage	115 ±5 volts AC 400 ±5 Hz 27.5 ±0.5 volts DC 26 ± 0.5 volts AC 400 ±5 Hz

4.5.1 Test Reading - Except where otherwise specified the transmitter can be tapped between 30 and 60 Hz with a maximum peak-to-peak amplitude of 0.002 inch before a test reading is taken.

4.5.2 Attitude - Unless otherwise specified, the transmitter shall be tested with the axis of the probe either horizontally or vertically mounted to simulate an aircraft installation.

4.6 Test Methods - The procedure specified herein shall be supplemented by a procedure outlined in detail by the supplier. This procedure shall state the exact conditions under which measurements are to be made. The procedure shall include details for test of all electrical, mechanical, and performance characteristics as specified for the particular transmitter, including provisions for indicating all variations in characteristics measured during the test procedure. This procedure is subject to the approval of the procuring activity.

4.6.1 Examination of Product - Each transmitter shall be weighed and examined carefully to determine conformance to the applicable drawings and to all the requirements of this specification not covered by tests.

4.6.2 Heater Power - Each heater of the transmitter shall be energized by applying rated voltage to the terminals and shall operate in still air at the rated power specified in MS28066.

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4.6.3 Mechanical Travel - With a line perpendicular to a line joining the reference dowel pins and bisecting the probe slots being considered mechanical center, the probe shall be rotated both clockwise and counterclockwise from this point to the mechanical stops Each stop shall limit the probe movement to one-half the nominal total angular probe travel, within 0 1 degree with respect to mechanical center.

4 6.4 Potentiometer Calibration - Position the probe to the center of its mechanical travel. Rotate the probe clockwise (looking at the probe end) to a point that is 45 percent of the nominal total mechanical travel from the mechanical center position and measure the resistance between the wiper and the low potential end. This resistance shall be 5.0 ±0.4 percent for wiper travel 100 percent of the total resistance or 9.5 ±0.4 percent for wiper travel 90 percent of total resistance as shown in MS28066 Rotate the probe counterclockwise to a point 45 percent from the mechanical center position and measure the resistance between the wiper and the high potential end. These values shall be within the tolerances specified above. Rotate the probe toward mechanical center and determine the electrical center (50 percent voltage ratio point) of each potentiometer Each electrical center point shall be within 0.1 degree of the center of the nominal mechanical travel of the probe.

4 6 5 Potentiometer Resistance - The electrical resistance of each potentiometer shall be measured between the pins at the electrical connector as specified in MS28066.

4.6.6 Potentiometer Noise - When the probe is oscillated over its mechanical travel at a frequency of approximately 10 cycles per minute and with rated voltage applied to the potentiometer under test the equivalent noise resistance measured between the wiper terminal and one end of the potentiometer at a maximum wiper current of 1.0 milliamp shall not exceed 100 ohms. Each wiper shall remain in contact with its winding for the full length of the mechanical travel of the probe. When the frequency of cycling the probe is increased to 60 cycles per minute the equivalent noise resistance shall not exceed 1000 ohms

4 6 7 Dielectric Strength - A potential of 500 volts (root mean square) alternating current at commercial frequency shall be applied between isolated circuits and between each isolated circuit and the case for a period of 5 seconds. There shall be no breakdown of insulation or any other permanent damage to the transmitter as a result of this test This test shall be applied to heater circuits when not common to the case The calculated leakage resistance shall exceed 5 megohms This potential shall not be applied more than 3 times on any one unit during unit tests.

4 6 8 Aerodynamic Alignment - The transmitter shall be mounted in a suitable wind tunnel so that the line joining the reference dowel pin centers is at right angles to the direction of airflow in the region where the probe is located When the airspeed is adjusted to 110 ±5 knots the electrical position output of each potentiometer and synchro shall be measured The 50 percent resistance point of each potentiometer and electrical synchro zero position of each synchro shall coincide with the probe aerodynamic center position within $\pm 0.1^{\circ}$ of probe rotation.

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4.6.9 Static Friction - Repeat the aerodynamic alignment test and displace the probe 0.3 ±0.05 degree in both directions from this position. After being released without impulse, the probe shall return to within 0.1 degree of its originally noted position as observed electrically.

4.6.10 Synchro Accuracy - The transmitter shall be mounted in a fixture which will provide means for rotationally positioning the probe at any angle within an accuracy of 0.05°. Readings of mechanical probe angle versus electrical angle of the synchro shall be taken at 5° increments of probe position over the entire angular travel. The electrical angular position of the synchro shall not differ from the probe position (multiplied by the nominal gear ratio) by more than 0.1° of probe rotation. The stator sequence shall be S3, S2, S1 for clockwise probe rotation for increasing angle of attack.

4.6.11 Magnetic Effect - The transmitter shall be energized and placed with its nearest part 12 inches from the center of an uncompensated short bar magnet compass in a horizontal magnetic field of 0.17 to 0.19 oersted intensity. When the transmitter is rotated 360 degrees about its two horizontal axes at each of the six 90 degree positions about the compass, the compass deflection shall not exceed 5 degrees from its reference position.

4.6.12 Potentiometer Linearity - The percentage of total potentiometer resistance shall be measured across the pins at the electrical connector for each wiper and the low end of its potentiometer, at approximately 5.0, 27.5, 50.0, 77.5 and 95.0 percent of nominal angular travel of the probe. The best calculated straight line through the observed percentage data shall not deviate from the five observed angles by more than 0.2 percent of the nominal angular travel of the probe.

4.6.13 Heating Element Endurance - With the transmitter mounted in a wind tunnel and the airspeed and air temperature adjusted to values such as to cause the thermostat to remain closed, rated voltage shall be applied to the heater circuits. The heaters shall be continuously operated for a period of 1 hour. The transmitter shall then be removed from the wind tunnel and the heater allowed to operate at standard conditions in still air for a period of 50 hours. No malfunction or damage to the transmitter shall result from this procedure.

4.6.14 Damping - The static friction test shall be repeated except that the probe shall be displaced approximately 25 percent of its total mechanical travel, and shall return to within 0.1 degree of its originally noted position without overshoot. Maximum flutter about this position, discounting airstream turbulence, shall be ±0.05 degree as observed electrically.

4.6.15 Time Constant - The static friction test shall again be repeated except that the probe shall be displaced approximately 3 degrees and the output of the potentiometer recorded. The time required for the probe displacement error to be reduced to 35 percent of its maximum value shall be 0.075 ±0.025 second as observed electrically.

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4.6.16 Low Speed Operation - The transmitter shall be mounted in a suitable wind tunnel so that the line joining the dowel pins is approximately right angle to the airstream. The airstream shall be adjusted to 50 knots. The transducer shall be mechanically displaced in the clockwise direction 5 degrees and released. The position of the potentiometer shall be read and recorded. The probe shall then be displaced in the opposite direction 5 degrees and released. The potentiometer position shall be read and recorded. The above test shall be repeated a minimum of 10 times. The difference between the RMS value of readings taken in one direction and the RMS value of readings taken in the other direction shall not exceed 0.5 degree. The maximum difference between corresponding readings when released from opposite directions shall not exceed 0.8 degree.

4.6.17 Vibration Error and Failure - Repeat the aerodynamic alignment test while vibrating the transmitter with the probe horizontal and then with the probe vertical in accordance with MIL-STD-810, Method 514, Procedure I, Figure 514-1, Curve D. The relationship of the airstream to the probe shall be such that the vibratory motion shall not cause the airstream to move across the slots thereby giving a false error signal indication. Connect the low potential end and the wiper arm of each potentiometer parallel to a 30,000 ohm resistive load and to the Y axis terminals of a suitable long persistence screen oscilloscope, and apply 10 volts DC across the potentiometer winding. The monitored error signal (potentiometer noise discounting airstream turbulence) shall not exceed 0.4 percent of the applied voltage. The synchro error shall not exceed 0.25 synchro degree. The heater circuits shall be monitored for continuity. After the vibration test, the transmitter shall be subjected to and meet all the requirements of the Individual Tests.

4.6.18 Pressure - A differential pressure of 15 psi shall be applied across the case of the transmitter enclosed in the area indicated in MS28066. No leakage shall be evident by observing that the pressure drop does not exceed 0.1 inch of mercury per minute.

4.6.19 Electromagnetic Interference - The transmitter shall be tested for conduction and radiation of and susceptibility to electromagnetic interference in accordance with MIL-STD-461 for Class ID equipment with each potentiometer, wiper, synchro and heater energized at rated voltage while the probe is turned 4 cycles per minute from stop-to-stop.

4.6.20 Torque - A torque of 45 to 50 pound-inches shall be applied to the probe without impact about the axis of probe rotation against each of the mechanical stops. The aerodynamic alignments shall be rechecked and shall be within specified limits.

4.6.20.1 Deflection of probe - A 150 pound load shall be applied at the tip of the probe at four 90° positions perpendicular with the probe axis, starting with the position halfway between the probe slots. There shall not be any permanent deformation or misalignment of the probe. The aerodynamic alignments shall be rechecked and shall be within specified limits.

4.6.21 De-icing - The transmitter shall be tested in an icing wind tunnel at an indicated airspeed of 350 ±10 knots, and an ambient temperature of -30 ±3° C. The transmitter shall be mounted in the wind tunnel as described

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in the Rain Test. The probe shall be coated with an ice layer approximately ¼ inch thick. Rated voltage shall then be applied to the heater circuit. The time required to clear the ice cap after the voltage is applied shall not exceed 2 minutes. Continue to operate for an additional 15 minutes at -30° C with 1.5 ±0.25 grams of water per cubic meter being introduced into the tunnel and the probe shall remain free of ice buildup. A potentiometer shall be monitored electrically to verify that ice does not form in the probe slots.

4.6.22 Low Temperature Exposure - The transmitter with no circuits energized shall be tested per MIL-STD-810, Method 502, Procedure I at -65 ±2°C for a period of 48 hours. The transmitter shall then be allowed to return to room temperature, after which it shall be subjected to and meet all the Individual Test requirements.

4.6.23 High Temperature Exposure - The transmitter with no circuits energized shall be tested per MIL-STD-810, Method 501, Procedure I at 93 ±2°C for a period of 48 hours. The transmitter shall then be allowed to return to room temperature, after which it shall be subjected to and meet all the Individual Test requirements. No damage or deterioration shall have occurred which will affect subsequent operation.

4.6.24 Shock - Subject the deenergized transmitter to MIL-STD-810, Method 516, Procedure I, Figure 516-2 Shock test. The transmitter shall then be subjected to and meet the Individual Test requirements.

4.6.25 Endurance - The probe of the transmitter shall be oscillated over the total range of angular travel at a frequency of 10 cycles per minute for a period of 100 hours. Rated voltage shall be applied to each potentiometer winding with each wiper connected to a 3,000 ohm resistive load. The transmitter heaters shall be energized for 10 minute periods after each 10 minute OFF period. After the test, the transmitter shall be subjected to and meet the requirements of the Individual Tests.

4.6.26 Humidity - The transmitter, with its connector contacts protected, shall be tested per MIL-STD-810, Method 507, Procedure I. At the conclusion of this test and within 1 hour after the transmitter has been removed from the chamber and the heaters have been energized for 10 minutes, it shall then be subjected to and meet all Individual Test requirements. There shall be no evidence of corrosion.

4.6.27 Fungus Resistance - The transmitter shall be subjected to MIL-STD-810, Method 508, Procedure I for an exposure period of 14 days. At the end of the test period, any evidence of fungus growth on the transmitter shall be cause for rejection.

4.6.28 Salt Fog - The transmitter, with its connector contacts protected, shall be subjected to MIL-STD-810, Method 509, Procedure I Salt Fog. No damage or deterioration shall have occurred which will affect subsequent operation. The transmitter shall then be subjected to and meet the Individual Test requirements.

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4.6.29 Dust - The transmitter with no circuits energized shall be mounted in a suitable chamber and subjected to MIL-STD-810, Method 510, Procedure I Dust. The mounting shall simulate a typical installation and the probe slots shall face upward and into the dust stream. At the conclusion of this test, the surface of the probe may be blown clear with an airstream provided the air does not remove or dislodge any internal dust. The transmitter shall be subjected to and meet the Individual Test requirement.

4.6.30 Rain - The transmitter shall be tested in a water spray wind tunnel at an indicated airspeed of 110 ±5 knots and at room temperature. Water spray equivalent to a heavy rain (1.0 ±0.1 inch per hour) shall be introduced into the wind stream. Only the probe shall be exposed to the wind stream. The mounting flange should be sealed to the tunnel wall to simulate installed conditions. The probe shall be horizontal with the slots facing upstream. A potentiometer shall be monitored electrically to verify that water does not block the probe slots. After a 15 minute exposure to the stream, the transmitter shall be subjected to and meet the Individual Test requirements.

4.6.31 Internal Examination - The case of the transmitter shall be removed and the mechanism including the electrical system shall be examined. Any damage or deterioration which could in any manner prevent the transmitter from meeting functional requirements during service life shall be cause for rejection.

4.7 Reliability Assurance Tests shall be conducted using MIL-STD-781 for Qualification Phase and Production Acceptance Phase tests. Each transmitter selected shall first have passed the Individual Tests. Test Level F of MIL-STD-781 shall be used for the 1,500 hour MTBF reliability tests while cycling the transmitter per the endurance test and performing the Individual Tests at 200-hour intervals and at the end of the tests.

4.7.1 Reliability Qualification Phase - Prior to approval of the transmitters, between 2 and 6 transmitters shall be tested as outlined in MIL-STD-781 "Qualification Phase of Production Reliability Tests" using the Accept-Reject Criteria for Test Plan III.

4.7.2 Reliability Production Acceptance Phase Tests - Between 2 and 6 transmitters shall be tested as outlined in MIL-STD-781 "Production Acceptance Phase of Production Reliability Tests" using the Accept-Reject Criteria for Test Plan V. Unless otherwise specified in the contract or order, no transmitters shall be shipped until an Accept decision is reached for each lot. The transmitters to be tested shall be selected at random throughout the lot by the cognizant government inspector.

4.7.3 Reliability Procedures and Reports shall be submitted in accordance with MIL-STD-781 using MIL-T-18303 format.

4.7.4 Longevity Tests shall be performed on two transmitters in accordance with Test Plan XXVIII and Test Level F of MIL-STD-781 for a minimum of 15 periods of endurance testing of 200 hours each with the Individual Tests performed after each period. Transmitters from the

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Reliability Qualification Phase testing may be used to accumulate the total testing time of 3,000 hours for each transmitter. Maintenance may be performed on each transmitter provided the total cost of servicing and replacement of parts throughout the longevity test does not exceed the "cost to repair" any transmitter as specified in the contract. (see 6.2)

4.8 Maintainability Demonstration shall be conducted in accord with MIL-STD-471, Test Method 1, Plan A_1 plus B_1 having a M_{ct} of 100 minutes and a $M_{max\ ct}$ of 300 minutes less the time to unseal and reseal the case. The tests shall be selected in advance and shall be approved by the procuring activity.

4.9 Reconditioning of Tested Transmitters shall be by the supplier replacing all wear or damaged items and made "good as new". All reworked transmitters resubmitted for acceptance shall be subjected to and meet all the Individual Test requirements.

4.10 Rejection and Retest - Rejected transmitters may be reworked to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning the rejection and the action taken to correct the defects found in the transmitter shall be furnished the government inspector. Failure of samples representing a lot shall be cause for the rejection of the lot.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the transmitters shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with MIL-STD-794 and MIL-I-81400.

6. NOTES

6.1 Intended Use - The transmitter covered by this specification is intended for use in aircraft as a means for determining the angle of air flow at the point of installation of the transmitter (local angle of attack, or sideslip) and to transmit electrical signals proportional to this angle to a pilot's angle of attack indicator and to other equipment requiring such signals.

6.2 Ordering Data - Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Quantity, MS part number, and interchangeability with the original supplier's stipulated transmitter for designated aircraft.

(c) Levels of packaging and packing, and markings desired.

(d) The laboratory that shall conduct sampling plan B tests.

(e) Reliability assurance tests, procedures, reports and size of a lot. (see 4.7)

(f) "Cost to repair" one transmitter. (see 4.7.4)

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6.3 Qualification - With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for the opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. Suppliers are directed to Provisions Governing Qualification SD-6 and are urged to arrange to have the product that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Air Systems Command, Department of the Navy, Washington, D. C. 20360, and information pertaining to qualification of products may be obtained from the activity.

6.4 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidary specification are in conflict, the following precedence applies:

(a) The Contract shall have precedence over any specification.

(b) This Specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(c) Referenced Specifications shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.5 Asterisks are not used to identify this revision change due to the extensive changes.

Project No. 6610-N003

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DEPARTMENT OF THE NAVY

Naval Air Systems Command
Washington, D.C. 20360

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DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines or reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION
MIL-[illegible] OF [illegible] MS-280DC

ORGANIZATION (Of submitter)

CITY AND STATE

CONTRACT NO

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4 REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)

DATE

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